Course Specification

Computer Graphics

(2026-2027)

1. Basic Information

Course Title (according to the bylaw)	**Computer Graphics**			
Course Code (according to the bylaw)	**IT311**			
Department/s participating in delivery of the course	**Information Technology**			
Number of credit hours/points of the course (according to the bylaw)	**Theoretical**	**Practical**	**Other (specify)**	**Total**
	2	**1**		**3**
Course Type	اجباري			
Academic level at which the course is taught	الفرقة/المستوي الثالث			
Academic Program	**Information Technology**			
Faculty/Institute	**Computers and Informatics**			
University/Academy	**Tanta University**			
Name of Course Coordinator	**Dr. Marwa Hassan Salama**			
Course Specification Approval Date	Click or tap to enter a date.			
Course Specification Approval (Attach the decision/minutes of the department /committee/council ....)				

2. Course Overview (Brief summary of scientific content)

This course introduces the fundamental concepts, algorithms and mathematics of computer graphics, starting from the pixel. It covers applications, graphics hardware and software, the graphics pipeline, raster and vector representations and color models; scan-conversion algorithms for lines, circles and ellipses (DDA and Bresenham/midpoint); parametric curves and splines (Hermite, Bezier and cardinal); polygon filling, flood fill and Gouraud shading; point, line and polygon clipping (Cohen-Sutherland and Sutherland-Hodgman); 2D and 3D geometric transformations using homogeneous coordinates and vector algebra; 3D object representation, the camera (view) transformation, orthographic, oblique and perspective projections and visibility; physically based animation with particle systems; and an introduction to virtual reality and stereoscopic rendering. All algorithms are traced through worked examples and implemented in Python alongside the original C++ listings.

3. Course Learning Outcomes CLOs

Matrix of course learning outcomes CLOs with program outcomes POs (NARS/ARS)

Program Outcomes (NARS/ARS) (according to the matrix in the program specs)		Course Learning Outcomes Upon completion of the course, the student will be able to:	
Code	**Text**	**Code**	**Text**
A2	Demonstrate strong knowledge of programming fundamentals, computer-based system construction, data structures and algorithms, software engineering techniques, and information retrieval.	**A2.1**	Translate the mathematical formulation of graphics algorithms (DDA, Bresenham/midpoint line, circle and ellipse algorithms, parametric curves and splines) into efficient code and trace them step by step.
A3	Show deeper understanding of advanced topics such as multimedia, computer and communication networks, data mining and knowledge discovery, information storage and retrieval, mobile communication systems, pattern recognition, artificial intelligence, cryptography, and network security.	**A3.1**	Explain advanced graphics topics including 3D viewing and projections, visible-surface determination, Gouraud shading, physically based particle simulation and stereoscopic rendering for virtual reality.
A6		**A6.1**	Implement rasterization, filling, clipping, transformation and projection algorithms in

Program Outcomes (NARS/ARS) (according to the matrix in the program specs)		Course Learning Outcomes Upon completion of the course, the student will be able to:	
Code	**Text**	**Code**	**Text**
	Understand and apply tools and techniques for the design and development of applications.		Python using a frame-buffer drawing runtime, and read their equivalent C++ listings.
		A6.2	Build animated and interactive graphics applications such as particle systems with numerical integration, collisions and springs, and head-tracked virtual reality views.
		A6.3	Describe the complete 2D and 3D graphics pipeline, from modeling and world coordinates through the view transformation, projection, clipping and rasterization to pixels on the screen.
B1	Analyze IT system problems, define goals, observe outcomes, reason effectively, and apply sound judgment.	**B1.1**	Analyze rendering problems (distorted, missing or incorrectly clipped objects) and use matrix and vector reasoning to correct transformation, viewing and projection errors.
B2	Identify attributes, components, relationships, patterns, key ideas, and errors in IT systems.	**B2.1**	Decompose complex scenes and 3D objects into geometric primitives (vertices, edges and polygon meshes) and organize them in hierarchical scene graphs.
C4	Use tools effectively for the implementation and documentation of computer-based systems.	**C4.1**	Develop a working graphics program from scratch that combines drawing, filling, clipping and transformation utilities, moving from equations to functional, tested code.
C6	Operate computing equipment efficiently considering both logical and physical properties.	**C6.1**	Document graphical algorithms and their implementations clearly (pseudocode, worked examples and commented code) so that the software is maintainable and verifiable.
D4	Exhibit appropriate numeracy skills in understanding and presenting cases involving a quantitative dimension.	**D4.1**	Perform and present the numerical calculations of computer graphics: tracing algorithm decision parameters, composing transformation matrices, computing projections, and evaluating frame-rate and latency requirements.
D7	Demonstrate an appreciation of the need to continue professional development in recognition of the requirement for life-long learning.	**D7.1**	Keep up with a rapidly evolving field (real-time rendering, simulation and VR hardware) through self-learning from textbooks, technical documentation and research papers.

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4. Teaching and Learning Methods

- Lectures
- Practical laboratories

- Worked examples traced step by step by hand
- Python programming exercises and mini-projects
- Interactive discussions
- Practice sheets for self-assessment (MCQ, fill-in-the-blank and problems)

Course Schedule

Number of the Week	Scientific content of the course (Course Topics)	Total Weekly Hours	Expected number of the Learning Hours			
			Theoretical teaching (lectures/discussion groups/ .....)	Training (Practical/Clinical/ .....)	Self-learning (Tasks/ Assignments / Projects/ ...)	Other (to be determined)
1	Introduction: Applications, Graphics Hardware and Software, the Graphics Pipeline and Color Models	4	2	2		
2	Line Drawing Algorithms (Direct, DDA and Bresenham)	4	2	2		
3	Circle and Ellipse Drawing Algorithms	4	2	2		
4	Parametric Curves and Splines (Hermite, Bezier and Cardinal)	4	2	2		
5	Filling Algorithms (Polygon Fill, Gouraud Shading and Flood Fill)	4	2	2		
6	Clipping Algorithms (Point, Cohen-Sutherland Line and Polygon Clipping)	4	2	2		
7	MidTerm	-	-	-		
8	2D Transformations, Homogeneous Coordinates and Scene Hierarchy	4	2	2		
9	3D Transformations and Vector Space	4	2	2		
10	3D Representation and the Camera (View Transformation)	4	2	2		
11	Projections (Orthographic, Oblique, Perspective), the 3D Viewing Pipeline and Visibility	4	2	2		
12	Particle Systems and Physical Simulation	4	2	2		
13	Virtual Reality: Stereoscopic Rendering, Head Tracking and Interaction	4	2	2		
14	Revision	4	2	2		

5. Methods of students' assessment

No.	Assessment Methods *	Assessment Timing (Week Number)	Marks/ Scores	Percentage of total course Marks
1	Midterm	7	25	25%
2	Final Written Exam	15	50	50%
3	Final Practical	13	15	15%
	Assignments / Project /Portfolio/ Logbook	8	10	10%

6. Learning Resources and Supportive Facilities *

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Learning resources (books, scientific references, etc.) *	**The main (essential) reference for the course** (must be written in full according to the scientific documentation method)	Salama, M. H. (Comp.). (2026). Computer Graphics (IT311): Rasterization, transformations, 3D viewing, simulation and virtual reality [Course book, with the companion booklet IT311 Practice Sheet Solutions]. Faculty of Computers and Information, Tanta University.
	Other References	- Hearn, D., Baker, M. P., & Carithers, W. R. (2011). Computer graphics with OpenGL (4th ed.). Pearson. - Marschner, S., & Shirley, P. (2016). Fundamentals of computer graphics (4th ed.). CRC Press. - Hughes, J. F., van Dam, A., McGuire, M., Sklar, D. F., Foley, J. D., Feiner, S. K., & Akeley, K. (2013). Computer graphics: Principles and practice (3rd ed.). Addison-Wesley Professional. - Akenine-Möller, T., Haines, E., & Hoffman, N. (2018). Real-time rendering (4th ed.). CRC Press. - LaValle, S. M. (2023). Virtual reality. Cambridge University Press.
	Electronic Sources (Links must be added)	- Eck, D. J. (2023). Introduction to computer graphics (Version 1.4). https://math.hws.edu/graphicsbook/ - LaValle, S. M. Virtual reality (online edition). https://lavalle.pl/vr/
	Learning Platforms (Links must be added)	-
	Other (to be mentioned)	-

Supportive facilities & equipment for teaching and learning *	**Devices/Instruments**	**Computer**
	Supplies	**-**
	Electronic Programs	**Python 3.8+ (standard library) and a C++ compiler for the original listings**
	Skill Labs/ Simulators	**-**
	Virtual Labs	**-**
	Other (to be mentioned)	**-**

Name and Signature **Course Coordinator**		**Name and Signature** **Program Coordinator**
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